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**UNIVERSITY OF
WATERLOO**



Design of Robust 635 LBF NOS-Ethanol Rocket Engine for Ground Testing

Prepared For:
Waterloo Rocketry

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2021 September 20

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Professor William Owen
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Dear Professor Owen,

This report, entitled "Design of Robust 635 LBF NOS-Ethanol Rocket Engine for Ground Testing", was written to fulfill my WKRPT 300 requirements.

The purpose of this report is to document the needs analysis and design of a test engine for the development of Waterloo Rocketry's liquid rocket engine. The liquid rocket engine is a continuation of early proof-of-concept work by Rajkumar, Jiang, Lin, Lai, and Kong in their 2020 FYDP project. Significant effort is being put towards development of the pintle injector and the test engine discussed in this report serves to characterize the pintle injector's performance. The overall liquid engine project aims to cumulate in a 10 000 ft apogee launch vehicle for the 2023 competition year.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I would like to thank Hamza Abu Abah, Aaron Leszkowiat, Abdulrahmen (ABM) Hussien, Stefan Martin, Samuel Street, Roman Kobets, Shirley Kong, and other Waterloo Rocketry team members for providing their advice and experience on this project. I received no other assistance.

Best regards,

A handwritten signature in black ink, appearing to read 'Zhen Bo Bian', with a stylized, cursive script.

Zhen Bo Bian
20763975
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Summary

Waterloo Rocketry is continuing development of a nitrous oxide-ethanol liquid rocket engine. Previous testing with an FYDP proof-of-concept engine demonstrated the risks of hard starts in liquid engine development as the engine was destroyed by a hard start. Further development of the pintle injector requires a robust test engine for short duration static fires to characterize performance.

In this project, a strong and inexpensive test engine is designed. Engine geometry is determined by Rocket Propulsion Analysis (RPA) software. RPA and Ansys FEA software are used to predict thermal behaviour of the engine for a 3 s static fire. The engine is overbuilt to withstand extreme pressures associated with hard starts to limit damage and delays in testing. The combustion chamber and nozzle are made of AISI 1020. A replaceable AISI 304 nozzle throat insert permits extreme throat wear over the course of a static fire. Bolted flange joints allow for modular assembly of the engine. The nozzle flange joint is designed to nominally fail at 1673 PSI combustion chamber pressure, acting as a pressure relief device.

1.0 Introduction

The Waterloo Rocketry student design team designs, builds, and launches high power sounding rockets [1]. Based out of the University of Waterloo, the team is composed primarily of undergraduate students, from a variety of programs, and competes in the annual Intercollegiate Rocket Engineering Competition (IREC) in Spaceport America, New Mexico. In past years, the team has competed with a student researched and designed (SRAD) hybrid bipropellant engine with a nominal apogee of 30 000 feet. In pursuit of fostering an environment for students to learn, apply, and improve engineering skills, the team has taken on the challenge of developing a liquid bipropellant rocket engine.

1.1 Fundamentals of Rocket Propulsion

Rocket engines apply Newton's Third Law of Motion to achieve propulsion. Newton's Third Law of Motion states that forces occur in equal and opposite pairs. By expelling mass away from the rocket, a thrust force acts on the rocket and propels it in the opposite direction [2]. Increasing the momentum of the propellant increases the thrust; momentum can be increased through increasing exhaust mass flow rate and exhaust velocity. Unlike an air-breathing jet engine, rockets carry all their propellants onboard. As a result, increasing thrust by increasing exhaust velocity is the preferred way to increase performance.

Chemical propulsion is the most common method of achieving thrust [2]. Chemical combustion is used to create high temperature, high pressure exhaust gasses in a combustion chamber before they are pushed through a nozzle and ejected to propel the rocket. Bipropellant engines are the predominant type and are typically classified as being a solid engine, liquid engine, or hybrid engine.

Solid engines utilize a solid propellant consisting of a premixed fuel and oxidizer [3]. Solid rockets are typically the simplest of the three types. Liquid engines utilize liquid propellants that are mixed prior to combustion. Liquid rockets are typically the most complex of the three types but yield the greatest performance ceiling. Hybrid engines inject a liquid propellant into the combustion chamber to react with a solid propellant. Hybrid rockets are an intermediary between the solid and liquid types in performance and complexity, while being safer to operate and store than either [3].

Liquid rockets need to store, mix, combust, and exhaust propellants. Typical designs include pressure vessels to store propellant, plumbing to transport propellant to the engine, an injector to mix propellants, a combustion chamber to contain the combustion reaction, and a nozzle to accelerate the combustion products.

Combustion generates extreme quantities of heat. The combustion chamber and nozzle must withstand extreme heat flux, which can exceed 16 kW/cm^2 [4], for the full burn duration. Forced convection is the predominant mode of heat transfer and radiation accounts for the remaining 5-35% of total heat transfer [4]. Figure 1 shows the typical heat transfer rate distribution in rocket engines. Heat flux is most intense at the throat of the nozzle. Thus, thermal analysis and cooling is an important aspect of rocket engine design. Methods of controlling intense heat transfer include regenerative, film, ablative, radiative, and conductive cooling [4].

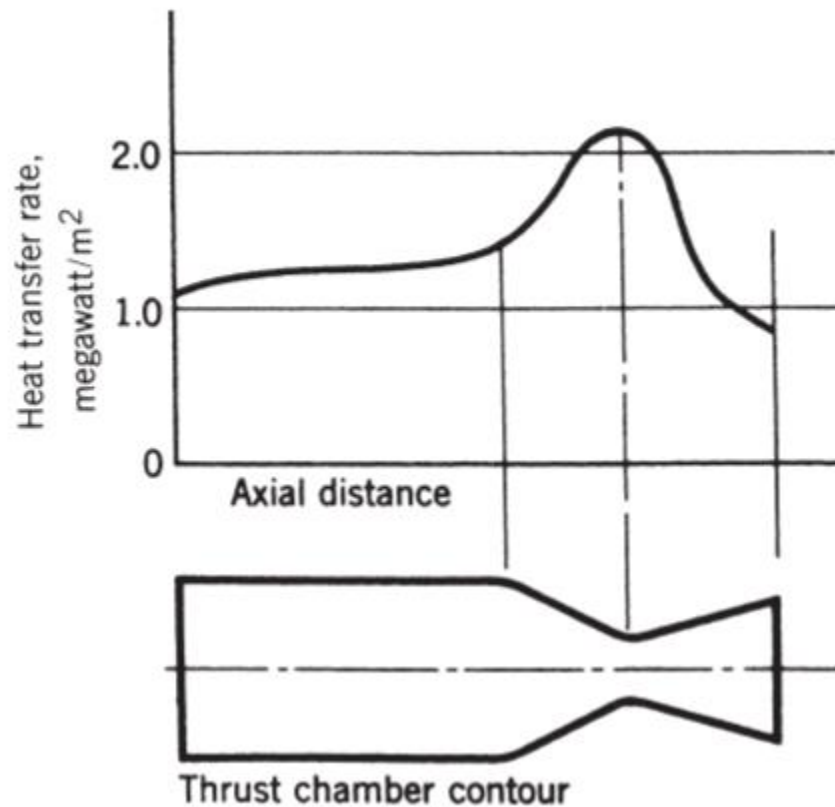


Figure 1: Typical rocket engine heat flux as a function of axial distance from the injector [4]

1.2 Motivation and Preceding Work

The motivation for this current project derives from the previous progress and experience of Rajkumar, Jiang, Lin, Lai, and Kong [5] in their fourth-year design project (FYDP) of developing a liquid rocket engine. A pintle injector was researched and developed for a nitrous oxide (NOS) and ethanol engine. A lightweight steel combustion chamber, G11 phenolic resin fibreglass-epoxy liner, and graphite nozzle were used in the design, similar to the hybrid engine developed by Waterloo Rocketry [5].

In the first static fire of this engine, a hard start occurred and destroyed the engine. Fragments of the engine are shown in Figure 2. Liquid engines are susceptible to intense pressure spikes during start-up transience, referred to as hard starts. Hard starts are expected with liquid engine development as they are a function of ignition and injection timing, among other factors. During initial liquid engine development, hard starts should be expected until parameters are refined to minimize pressure spikes. The hard start that destroyed the FYDP engine exceeded an overpressure of 2700 psi based on the observed bolt shear failure mode [5].



Figure 2: FYDP engine injector bulkhead with pintle after hard start, note the sheared bolts [6]

Similar liquid engine development projects by others have experienced their own challenges or have actively designed to minimize the negative impacts of these challenges. During testing of a liquid oxygen and gasoline engine, Florián, Urrego, and Rojas experienced catastrophic burn through of the combustion chamber walls due to the use of AISI 1020 steel in high temperature oxygen service [7]. By utilizing a high factor of safety (FoS) copper engine, Youngblood was able to continue a static fire, even after a hard start [8]. Similarly, Austin and Heister were able to resume their hypergolic rocket engine test campaign with little delay after a hard start by designing bolted flange joints to function as a pressure relief device [9].

As such, a robust engine for preliminary testing and characterization of Waterloo Rocketry's liquid engine pintle injector is desired. An engine capable of safely containing hard starts that also mitigates the risks of thermal failure facilitates a high test tempo and iterative design approach to the pintle injector.

1.3 Engine Parameters

The Waterloo Rocketry bipropellant liquid engine is a continuation of the FYDP liquid engine project. This engine continues to use NOS and ethanol as the oxidizer and fuel, respectively. In flight configuration, this engine will have a nominal apogee of 10 000 feet. The engine has a chamber pressure of 300 PSI and an expected thrust of 635 lbf. The combustion chamber inner diameter is 5", compared to the 3.5" inner diameter combustion chamber used in the FYDP design [5]. As such, the overall length of the combustion chamber is less. For simplicity and familiarity, a linear approximation bell nozzle profile with 30° converging and 15° diverging half angles is used as was on the FYDP engine [5].

The engine is designed to use a pintle injector with mass flow rates of 1.0 kg/s and 0.2 kg/s for the oxidizer and fuel, respectively. Pintle injectors create internal combustion flows with recirculation zones, illustrated in Figure 3. This aspect of combustion will have impacts on the required length of combustion chamber and thermal analysis.

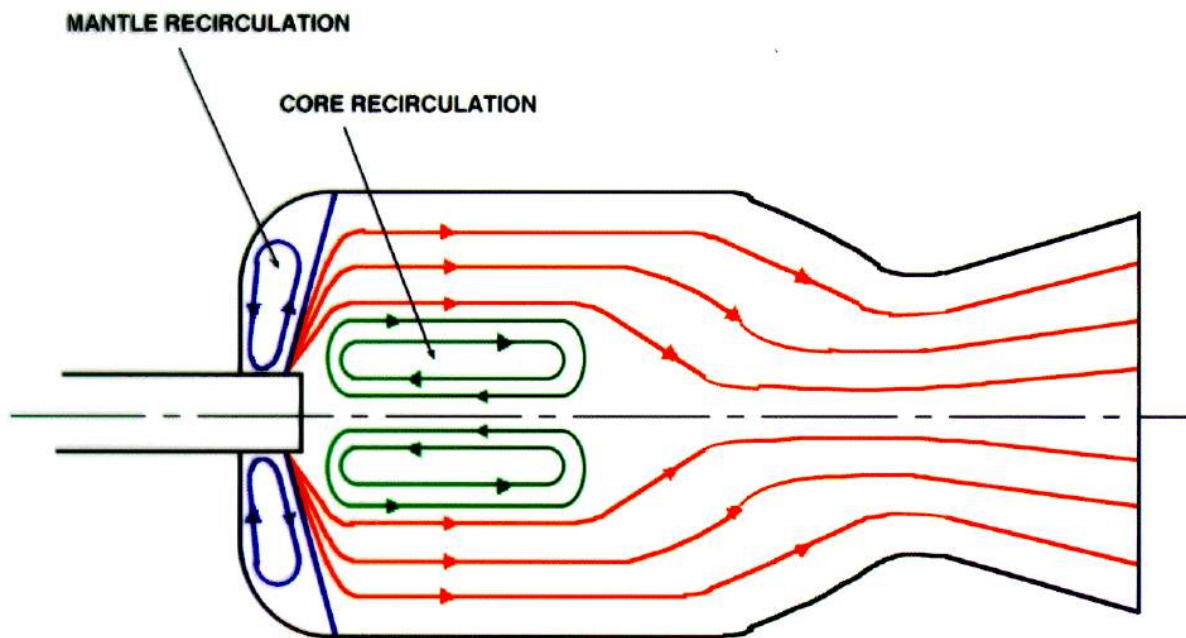


Figure 3: Internal combustion flows created by pintle injectors [10]

The complete engine will be tested at the Waterloo Rocketry Columbia Lake test site "Texas". The design will need to attach to the modular test stand used by Waterloo Rocketry.

1.4 Long-Term Development and Direction

Waterloo Rocketry has pursued the development of a liquid rocket engine in pursuit of higher performing rockets and greater technical challenges. Liquid rocket engines are able to achieve greater specific impulse compared to hybrid rocket engines of similar size [5]. This allows for higher performing, more efficient launch vehicles in the future. Hybrid rocket engines do not scale as effectively as liquid rocket engines due to the use of a solid fuel grain that occupies the combustion chamber. Liquid rocket engines also bring many new technical challenges to foster learning and skill development for the Waterloo Rocketry team, satisfying the core team objective of developing student engineering skills [1].

The high-level objective of the liquid rocket engine project is to produce a 10 000 feet apogee launch vehicle for flight during the 2023 competition year. After the 10 000 feet apogee rocket is successfully flown, a 30 000 feet rocket is the expected next step.

This robust engine project plays a critical role in early development of the liquid engine: it provides a safe and forgiving platform for rapid and inexpensive testing of the pintle injector. This engine will also allow for characterization of other parameters critical to the design of a flight engine, including thermal behaviour. This project leads into flight engine design and cooling system design.

1.5 Objective

The objective of this project is to design, manufacture, assemble, and test a robust rocket engine for development of a pintle injector. The final deliverable is a complete and tested thrust chamber. The engine is expected to facilitate testing and development of the pintle injector. The scope includes the combustion chamber, the nozzle, injector attachment, and engine test stand attachment.

This report is created to document the state of the project and major design decisions made up to September 2021. The next steps of this project are detailed at the end of this report.

2.0 Problem Definition

Waterloo Rocketry needs a robust ground test engine for preliminary development of a liquid NOS-ethanol bipropellant pintle injector for a 10 000 feet apogee rocket. The engine needs to withstand the thermal extremes of a 3 second long static fire and be reusable after hard starts. The engine must facilitate a high test tempo and be inexpensive to manufacture and operate.

2.1 Applicable Standards

Applicable standards are listed in Table 1, in order of descending precedence. The SA Cup Rules and Regulations and Design, Test, & Evaluation Guide hold the same level of precedence.

Table 1: Applicable standards, in descending precedence

STANDARD	JURISDICTION
SA Cup Rules and Regulations [11]	Experimental Sounding Rocket Association
Design, Test, & Evaluation Guide [12]	Launch Canada Challenge
Waterloo Rocketry CAD Standard [13]	Waterloo Rocketry

2.2 Timeline

This project started in May of 2020. However, due to COVID-19, significant delays have occurred. During the first year of development, significant research was conducted into liquid engine cooling systems and materials for high temperature service. The 10 000 feet apogee liquid rocket that this project is working towards is expected to launch in the summer of 2023. A series of rigorous test campaigns will take place before competition to ensure a successful flight.

Design lock will occur in early October 2021. Manufacturing and assembly follow design lock. Testing and verification of design will occur in October and November 2021. The engine will need to be ready for short duration static fires with the pintle injector by late November 2021.

2.3 Requirements, Constraints, and Criteria

Requirements, constraints, and criteria are centered around ensuring operation through a hard start and economy of manufacture and operation.

2.3.1 Requirements

Requirements that the design must meet are as follow:

- Sustain a short burn of at least 3 seconds at nominal burn conditions
- Contain hard start pressure up to approximately 2000 PSI
- Fail safely when hard start pressure exceeds approximately 2000 PSI
- Sustain a minimum of 10 static fires

Short burns of 3 seconds are estimated to be sufficient to qualify injector performance [9]. The scope of this engine is limited to short duration burns to simplify cooling needs of the design. 2000 PSI is set as the rough upper limit for hard start pressure based on regulator and check valve availability from McMaster-Carr [14] and to avoid extreme pressure seal design. Safe failure above the design pressure is required as an immediate pressure relief mechanism. A minimum of 10 static fires is estimated to be required for injector testing.

2.3.2 Constraints

Constraints that the design must adhere to are as follow:

- Be suitable for testing at the Columbia Lake test site, “Texas”
- Accommodate the injection and ignition systems

Static fire testing takes place at the Waterloo Rocketry Columbia Lake test site, so this engine will need to be operable in the conditions expected at an outdoor facility with similar considerations for resources available. Key to the objective of the liquid engine development campaign, the engine must accommodate and conform to the injection and ignition systems.

2.3.3 Criteria

Criteria that apply to the design are listed in descending precedence as follow:

1. Minimize damage to plumbing and nearby equipment during expected failure
2. Minimize the time required to replace injector components and consumables
3. Minimize the manufacturing time and cost of consumable parts
4. Minimize the quantity and usage rate of consumable parts
5. Minimize the cost of permanent parts
6. Maximize the simplicity of manufacture and assembly of permanent parts
7. Minimize the need for additional test infrastructure

Damage to plumbing and nearby equipment risks significant test delays and expenses due to high component cost, long component lead time, and intensive assembly procedure. To facilitate iterative pintle injector development, the rate at which different injector variations can be installed and tested should be maximized to allow a greater number of tests to occur within a test session. To facilitate repeated testing, the consumable parts should be inexpensive and readily producible, and the number of consumable parts and their replacement rate minimized to further decrease operational expenses. The cost and simplicity of permanent parts should be minimized to meet the limited budget allotted to this project. The logistical burden of testing at Texas is already significant. By limiting the need for additional test infrastructure, increases to the logistical burden are reduced.

3.0 Engine Design

This section of the report covers the detail design of the engine. An overview is provided to discuss the main features of the design, followed by an explanation of the two system performance cases used to drive the design. The thermal study and materials selection is then discussed, followed by detailed design discussions of the combustion chamber, nozzle, and bolted flange joints.

3.1 Design Overview

The engine consists of a combustion chamber, nozzle, and nozzle throat insert (insert), shown in Figure 4. The nozzle is fixed to one end of the combustion chamber and the injector (not shown) is attached to the opposing end. The insert is a sacrificial part designed to bear the intense heat transfer at the throat of the nozzle. The insert is retained by a machined shoulder on the nozzle and a bolted flange. Bolted flange joints are used to join the components together and sealing is achieved by face sealing O-rings. The combustion chamber and nozzle are of AISI 1020 while the insert is of AISI 304.

This design was chosen for robustness, simplicity, and modularity. Thick walls and flanges prevent yielding during a hard start such that the combustion chamber and nozzle may be reused after hard starts. Bolted flange joints can hold high pressures and can be designed to fail at a set pressure. By failing at a designated pressure, the bolted flange joint acts like a large burst relief disk to prevent excessive pressure buildup. As such, the combustion chamber and nozzle only need to be designed to handle pressures up to the failure pressure of the bolted flange joint. The use of pins in shear was considered. However, bearing failure would be expected during a hard start, thus elongating the pin holes and being non conducive to reusability. In addition, the bolted flange joint allows for different nozzles and lengths of combustion chamber to be tested. This gives the design flexibility for further use beyond preliminary testing of the pintle injector.

Heat sink cooling is chosen for simplicity and ruggedness. The replaceable throat insert simplifies design by eliminating the need for active cooling of the nozzle and allows for the use of inexpensive and machinable materials. Without the use of expensive or difficult to manufacture specialty materials such as ceramics, high temperature alloys (e.g., Inconel), or graphite, erosion of the nozzle throat is unavoidable. A replaceable insert allows for replacement of only the high wear throat region rather than the entire nozzle. This saves significant material costs as only a small part needs to be replaced on a regular basis instead of the entire nozzle.

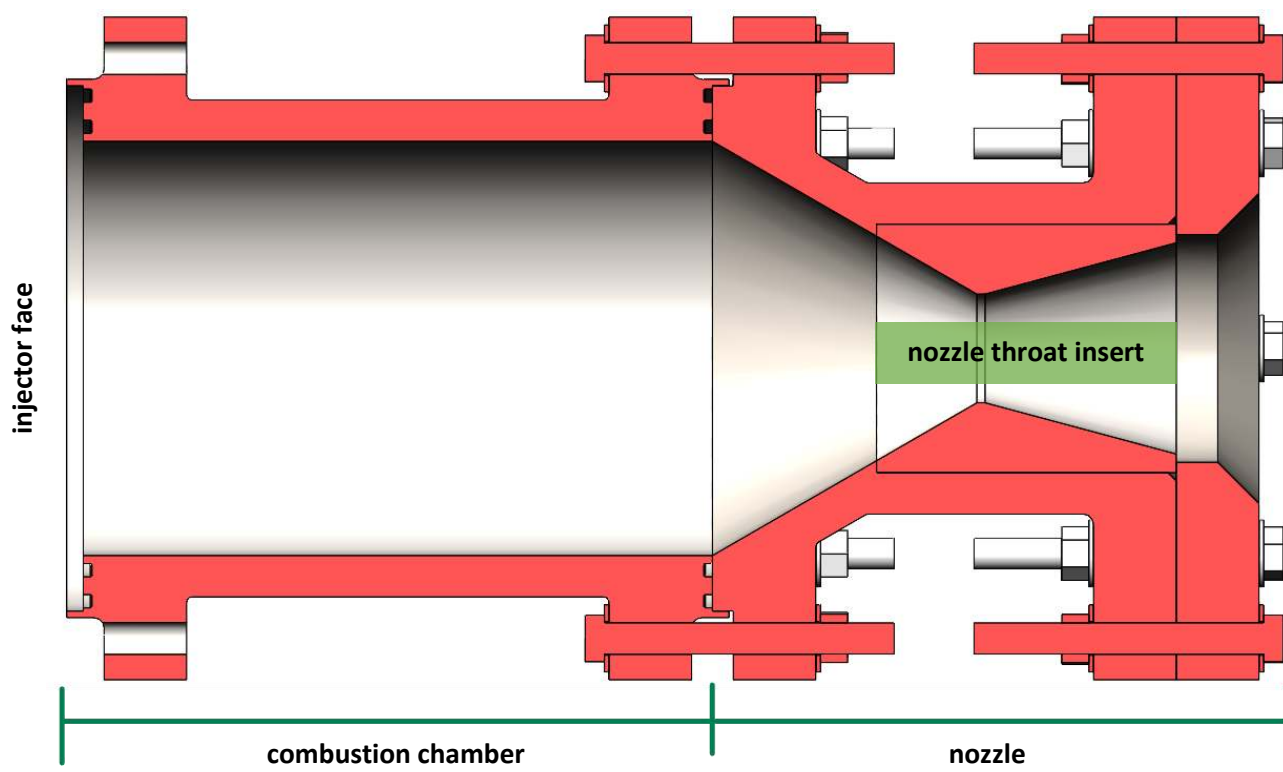
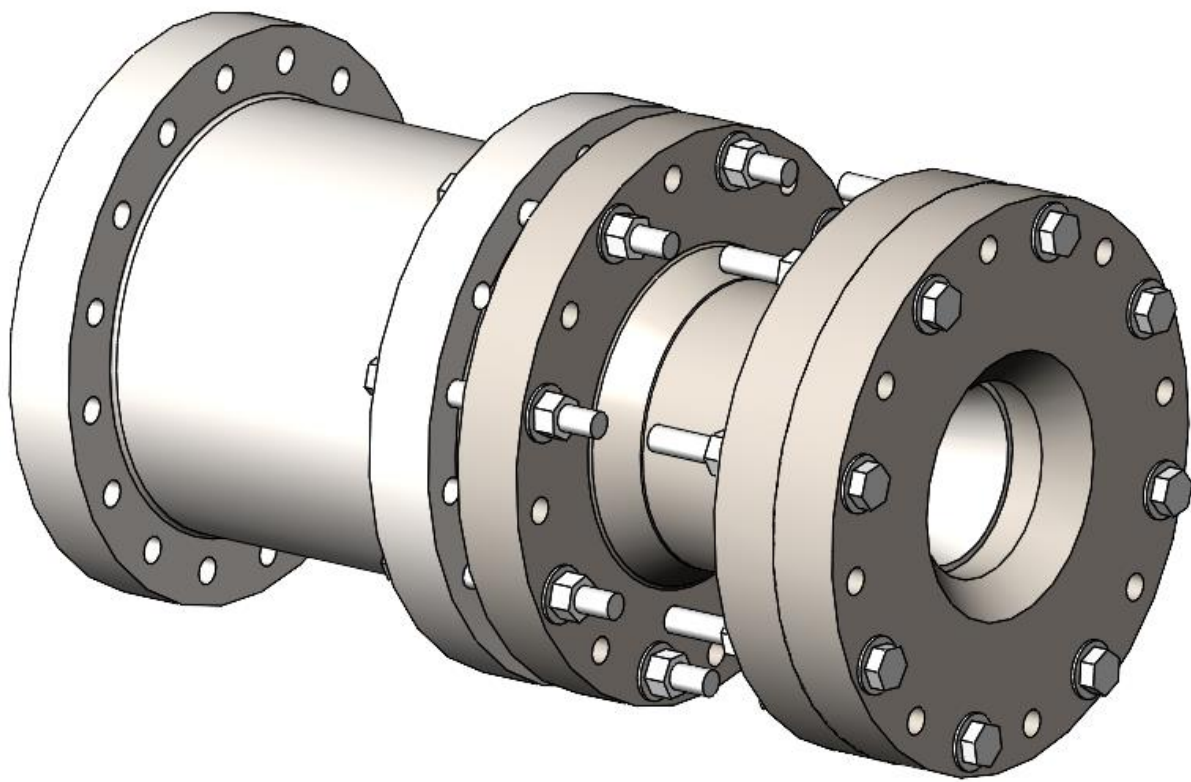


Figure 4: Layout of engine components, isometric view top, section view bottom

3.2 System Performance Cases

Two system performance cases are used to drive the design of the engine: 1) a nominal 3 second static fire and 2) a hard start.

A nominal 3 second static fire entails the case of an expected static fire. Nominal combustion is a 3 second burn at 300 PSI, starting from ambient temperature and pressure (20 °C and 1 atm). 3 seconds was selected as an appropriate amount of time to capture start up transience and steady state combustion to characterize injector performance. Results from a similar test campaign with over 150 firings on a hypergolic bipropellant, pintle injector, liquid engine, conducted by Austin and Heister [9], indicate that a 3 s burn is sufficient to characterize engine performance by the plateau in chamber pressure after 0.5 s of a typical burn shown in Figure 5. To simplify analysis, steady state combustion characteristics are assumed for the entire 3 s burn duration. A minimum FoS of 6 is required for this case. The combustion chamber, nozzle, and insert are assumed to be in an annealed state as a result of previous burns.

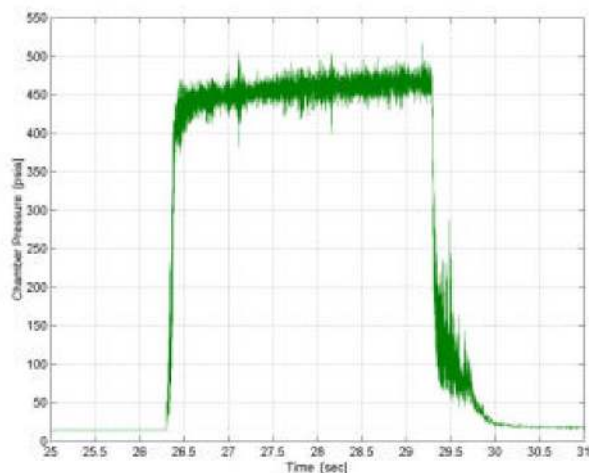


Figure 5: Typical chamber pressure variation with time for a 3 s burn test campaign by Austin and Heister [9]

A hard start entails an overpressure exceeding 2000 PSI immediately following ignition while the engine is still at ambient temperature. Hard starts occur at the start of a burn before steady state combustion is achieved. Heat transfer leading up to a hard start is ignored due to the short time duration between ignition and overpressure. As such, analysis is performed assuming the components remain at ambient temperature. The rationale for setting 2000 PSI as the approximate overpressure limit is discussed in section 2.3.1 *Requirements*. A minimum FoS of 3 is required for the combustion chamber, nozzle, and insert while an FoS of 0.8-1.2 is required for components intended to fail in the

hard start. Like the nominal burn case, the combustion chamber, nozzle, and insert are assumed to be in an annealed state because of previous burns.

3.3 Thermal Analysis and Materials Selection

Thermal analysis and appropriate materials selection are critical for preventing mechanical failure at elevated temperatures and thermal material failure over the course of a 3 s burn. Rocket engines are known for their extreme thermal loads so estimating the temperature gradient allows for approximating changes in material properties with temperature, thermal stresses, the rough order of magnitude of thermal fatigue life, and the risk of thermal material failure. The mechanical properties of metals can vary significantly with extreme temperature changes so design calculations must account for the increase in temperature over the course of a burn. The high heat flux of rocket engines develops significant temperature gradients within parts resulting in thermal expansion stresses. Thermal fatigue worsens as the difference in cyclic temperatures increase, resulting in significantly accelerated mechanical failure [15]; estimating the rough order of magnitude of thermal fatigue life can set a conservative upper limit on test cycle life. Material ignition and subsequent thermal failure are possible due to the hot combustion products and the presence of oxidizer, as demonstrated by an engine burn-through in a LOX-gasoline engine [7].

The result of this thermal analysis also determines if additional cooling measures are required. The least complex and most economical cooling option is desired. As such, heat sink cooling is the primary method of cooling as it does not complicate the design, nor introduce additional infrastructure. For short duration burns with no mass budget, a heat sink cooled engine is appropriate [4]. The G11 ablative liner used on the FYDP engine can be used if additional cooling is required by evidence of excessive wear and excessively high temperature readings. Active cooling methods such as film cooling and regenerative cooling are not pursued due to their cost and complexity.

AISI 1020 or similar mild steel is preferred for its low cost and machinability. The combustion chamber and nozzle are made of AISI 1020. AISI 304 is used where temperatures are too high for AISI 1020. AISI 304 is the preferred high temperature material due to its relatively low cost, availability, and machinability. AISI 304 is used for the insert.

Thermal analysis is conducted in Ansys FEA software with forced convection coefficients and hot gas-side temperatures determined by Rocket Propulsion Analysis software (RPA). The parameters specified in section 1.3 *Engine Parameters* are used as inputs in RPA. RPA calculates engine geometry, convection coefficient, and hot gas-side temperature results; these results are included in Appendix A.

The RPA results are used as input parameters in an Ansys transient thermal analysis project. An axisymmetric analysis with simplified geometry is utilized to decrease solver run time. The mesh is shown in Figure 6. The mesh consists of 7671 nodes and 2350 elements. Element edge sizing of 0.001 m is applied to the wetted surfaces to better capture the thermal gradient at the wetted surfaces. Granta EduPack temperature dependent material properties for AISI 1020 are applied to the combustion chamber and nozzle, and AISI 304 applied to the nozzle insert.

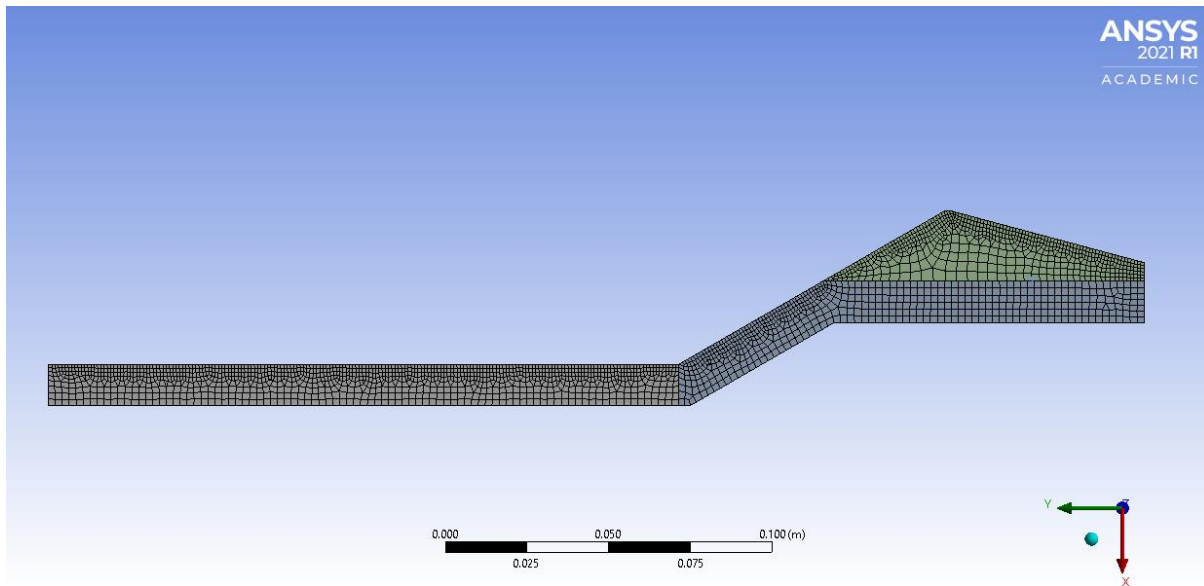


Figure 6: Ansys mesh with simplified engine geometry

The temperature results plot for a 3 s burn assuming constant steady state combustion is shown in Figure 7. Maximum temperature in the combustion chamber is 104 °C. Maximum temperature in the nozzle is 256 °C. Maximum temperature in the insert is 1326 °C. In Figure 7, it is observed that most of the insert remains below 500 °C and that the temperature extremes in the insert are isolated to the wetted surface, particularly near the throat.

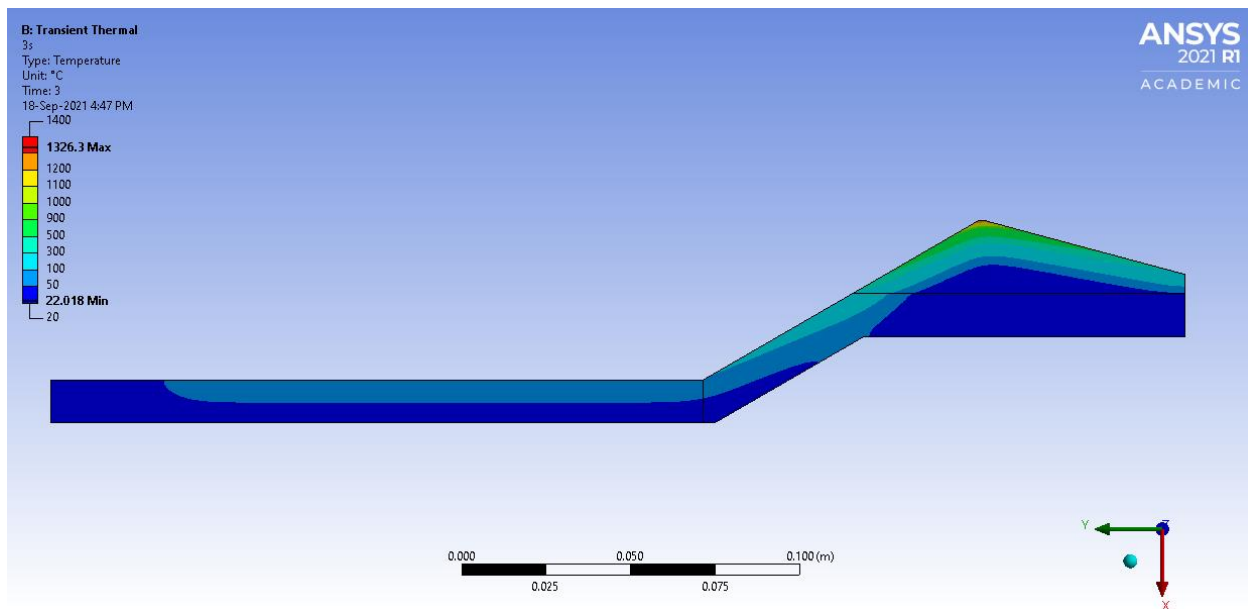


Figure 7: Ansys temperature results plot for a 3 s static fire with nominal steady state combustion

There are limitations with this thermal analysis. The RPA convection heat transfer parameters assume a homogenous combustion product moving axially towards the exhaust and do not factor in the internal recirculation patterns associated with pintle injectors [16]. Therefore, the results of this thermal analysis are an estimation. Thorough temperature measurements taken during static fires are required for further thermal modelling. To account for this variability, design calculations are performed with material properties assuming 500 °C bulk temperature. This helps account for hot spots and gross increases in temperature.

Transient structural analysis in Ansys is used to estimate the thermal stresses of a 3 s burn. The temperature results from the preceding thermal analysis are input as a thermal load in this analysis. A 300 PSI pressure is applied to all wetted surfaces to add the stress expected during a burn. The von-Mises yield stress FoS plot is shown in Figure 8. Most of the system has a FoS > 1. Limited portions have a FoS < 1. However, as this is a FoS for yield, the design is not expected to fail. Minor, localized permanent deformation may occur, but this is acceptable as it does not occur near critical sealing surfaces.

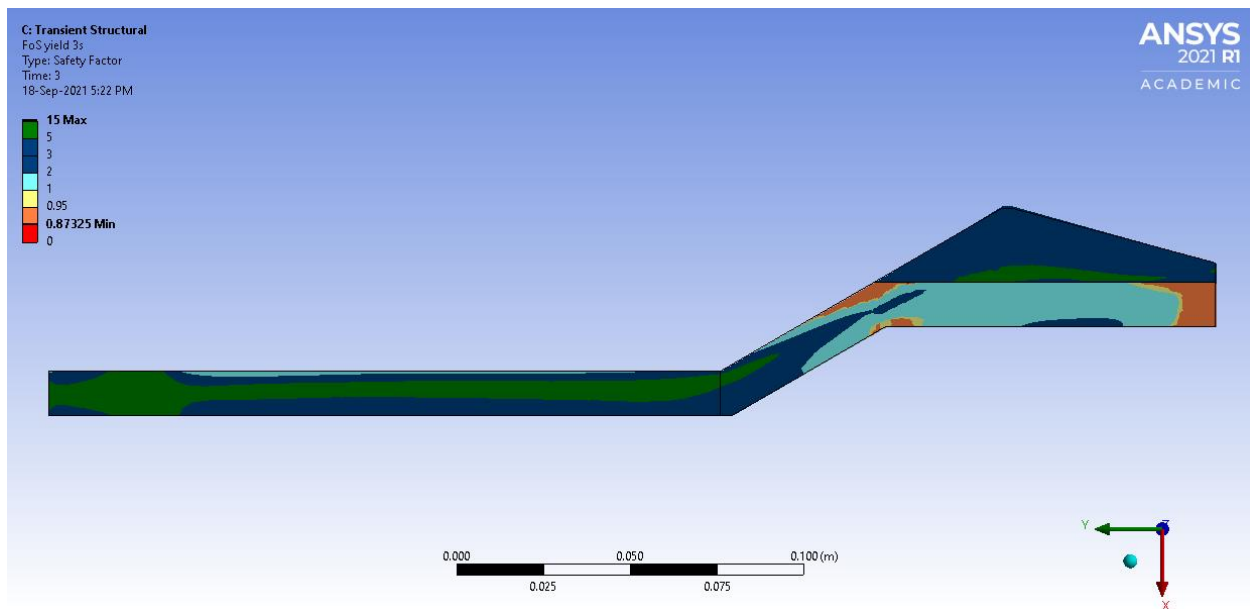


Figure 8: Von-Mises yield stress FoS plot for a 3 s burn with steady state combustion

Aoi, Marumiya, and Ebara have shown that low carbon steel (SB410) of 12 mm and 24 mm thickness fail after 24 and 23 thermal cycles, respectively, when heated to 500 °C and quenched with tap water [17]. As 500 °C represents a conservative max temperature assumption, the combustion chamber and nozzle should be able to meet the 10 test cycle requirement, especially since the engine will not be quenched after a burn. Arora has demonstrated that ½" rod AISI 304 has a thermal fatigue life of approximately 75 cycles when cycled between 40 °C and 1000 °C, with air cooling [15]. This high thermal fatigue life combined with the sacrificial intent of the insert mean that the insert can be reused if wear is negligible.

Metals can ignite and rapidly fail in high temperature, oxygen rich environments. At high temperatures, the protective oxide layer of metals fails. If this failure occurs in an oxygen rich environment, the metal can ignite and burn, and in certain cases will self-sustain combustion [18]. The ignition temperature of metal is dependent on sample geometry, oxygen partial pressure, and heat source, among other factors [19]. There lacks comprehensive research on NOS compatibility with metals for high temperature service. Current literature and research are limited to high temperature and high pressure oxygen service. This information can be used to set a conservative upper temperature limit for AISI 1020 and AISI 304.

Bolobov observed ignition of steel in atmospheres of 0.2 – 20 MPa oxygen at temperatures between 1250 °C and 1340 °C [20]. *Safety Design for Space Systems* notes an ignition temperature range of 1227-1277 °C for mild steel in atmospheric air [21]. Muller observed ignition of mild steel in 2-200 bar

oxygen at 1377 °C [19]. Nguyen and Branch observed ignition of AISI 1018 in 0.084-6.895 MPa of oxygen below its melting range of 1389-1412 °C [22]. Tench, Roder, and Clark observed ignition of carbon steel at 1097 °C [18]. It is noted that Tench, Roder, and Clark's ignition temperature is comparatively low. Thus, the maximum service temperature of AISI 1020 in this design is set at 1227 °C. Thus, assuming a conservative bulk temperature of 500 °C in the combustion chamber and nozzle, bulk thermal failure is not expected.

Bolobov, Berezin, Drozhzhin, and Shteinberg observed ignition of austenitic stainless steel in atmospheres of 0.2-20 MPa oxygen at temperatures between 1330 °C and 1420 °C but did not observe ignition in air before melting [23]. Muller observed ignition of austenitic stainless steel in 2-200 bar oxygen at 1627 °C [19]. Tench, Roder, and Clark observed ignition of austenitic stainless steels between 1127 °C and 1397 °C [18]. Like with carbon steel, it is observed that Tench, Roder, and Clark's ignition temperature for stainless steel is comparatively low. Thus, the maximum service temperature of AISI 304 in this design is set at 1330 °C. The temperature of the wetted surface of the insert in the nozzle throat region approaches 1330 °C but this is acceptable as the insert is a sacrificial component and significant wear is expected regardless.

Based on this analysis, additional cooling is not required.

3.4 Combustion Chamber Design

The combustion chamber dimensions are driven by the RPA results, included in Appendix A. The combustion chamber is made of annealed AISI 1020. The inner diameter is 5", the walls are 0.5" thick, and the chamber is 7.6" long. The RPA results specify a 5.9" long combustion chamber but this value does not account for the pintle injector. As mixing occurs at the tip of the pintle injector, which protrudes into the chamber, the pintle tip is set to be 5.9" from the nozzle. Thus, the entire chamber is 7.6" long.

Thick-wall pressure vessel calculations and the von-Mises yield criterion are used to determine the FoS for nominal burn and hard start. The combustion chamber is conservatively simplified as a closed, cylindrical pressure vessel. The summary of the results is listed in Table 2. Detailed calculations are included in Appendix B.

Table 2: Combustion chamber factor of safety

System Performance Case	Internal FoS	External FoS
Nominal 3 s burn	13.6	19.7
Hard start	3.5	5.0

3.5 Nozzle and Insert Design

The nozzle and insert are designed to be robust, economical, and modular. The wetted geometry is set by the RPA results, included in Appendix A. A linear approximation de Laval Nozzle, similar to the geometry used by Waterloo Rocketry, is chosen for manufacturability on a manual lathe. A section view of the nozzle assembly is shown in Figure 9. The nozzle assembly consists of the nozzle, the throat insert, and the retaining flange. The insert is retained in place by a shoulder in the nozzle and by the retaining flange that is fastened to the nozzle by eight 3/8"-16 grade 5 bolts and nuts.

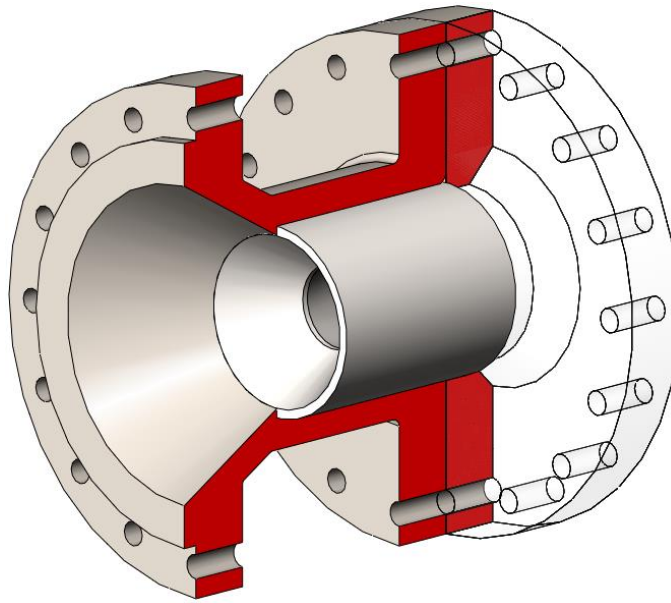


Figure 9: Nozzle assembly section view, nuts and bolts hidden in view

Similar to the combustion chamber, the nozzle is made of 0.5" thick AISI 1020 and the retaining flange is made of 1" thick AISI 1020. The nozzle throat is designed to be turned from 3" diameter AISI 304 round stock. This size is chosen for its availability and low cost from the E3 Engineering Machine Shop. Additionally, the 3" diameter insert completely encompasses the diverging section of the nozzle; an ideal insert would also encompass the converging portion to make all wetted surfaces in the nozzle replaceable.

Through the use of a replaceable throat insert, the main body of the nozzle can be saved from intense wear and heating to allow reuse. Only the small insert needs to be replaced after it has accumulated sufficient wear to negatively impact test results.

Bolted flange joints serve as a robust and inherently modular way of attaching parts together. This connection type allows different nozzle body materials be used if testing shows that heating is more severe than expected. Additionally, by use of a machined shoulder and bolted retaining flange, higher temperature materials may be readily utilized for the insert if needed, such as graphite.

The machined shoulder also serves to block a parallel path for hot gas flow. Manufacturing and assembly tolerances will require a small diametral gap between the nozzle and the flange, and the shoulder blocks direct access to this gap from the axial bulk flow of combustion gasses.

The retaining flange and its bolt pattern is sized by the two system performance cases. The factors of safety for the insert shearing through the retaining flange is detailed in Table 3. The outer diameter is taken as 3", the chamber pressure is assumed constant, and the throat is neglected for simplicity. The nozzle half of the retaining flange joint will have a higher FoS than those shown in Table 3 as the shear area is greater. Flexure of the retaining flange joint during overpressure is negligible due to the extreme thickness of flanges. Therefore, the nozzle is securely fixed within the nozzle. Retaining flange bolt preload, preload % yield strength, and hard start % ultimate tensile strength calculations are summarized in Table 4. The axial bolt preload is set such that the retaining flanges do not separate during a hard start. Detailed calculations are included in Appendix B.

Table 3: Insert shear through retaining flange FoS

System Performance Case	FoS
Nominal 3 s burn	59.5
Hard start	15.2

Table 4: Bolt pattern preload, preload % yield strength, and hard start % ultimate tensile strength

Per Bolt Preload	Preload % Yield Strength	Hard Start % Ultimate Tensile Strength
1767 lbf	24.9 %	38.2 %

Alternative arrangements of retaining the nozzle with radial patterns of pins or shoulder screws were considered but deemed insufficiently robust and unnecessarily difficult to manufacture. Such a concept is illustrated in Figure 10. The pin holes would sustain bearing failure during a hard start, preventing reuse of the nozzle. In addition, the architecture is difficult to over strengthen due to the limited circumference of the nozzle, risking premature shear failure of insert-retaining fasteners before

intended tensile failure of bolted flange joint fasteners. Additionally, the manufacturing tolerances required for aligning a radial pattern of dowel pin holes is impractical for the disposable insert.

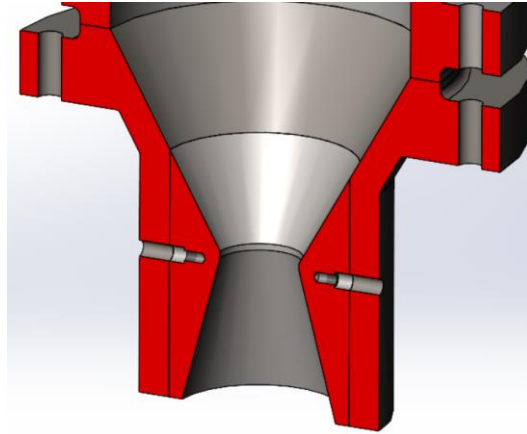


Figure 10: Radial shoulder screw insert retention concept, assessed to be impractical

3.6 Bolted Flange Joint Design

Bolted flange joints are chosen for modularity, robustness, and strength. A section view of the flange joint is shown in Figure 11. Flanges are 8" diameter, 1" thick, and of the same AISI 1020 as the combustion chamber and nozzle. The flanges have a 16-bolt hole pattern on a 7" bolt circle for 3/8" diameter fasteners. The same design of flange is applied universally across the engine for modularity and symmetry. Face sealing O-rings are selected for joint sealing because they are simple to implement and have a proven track record in similar applications [8, 9].

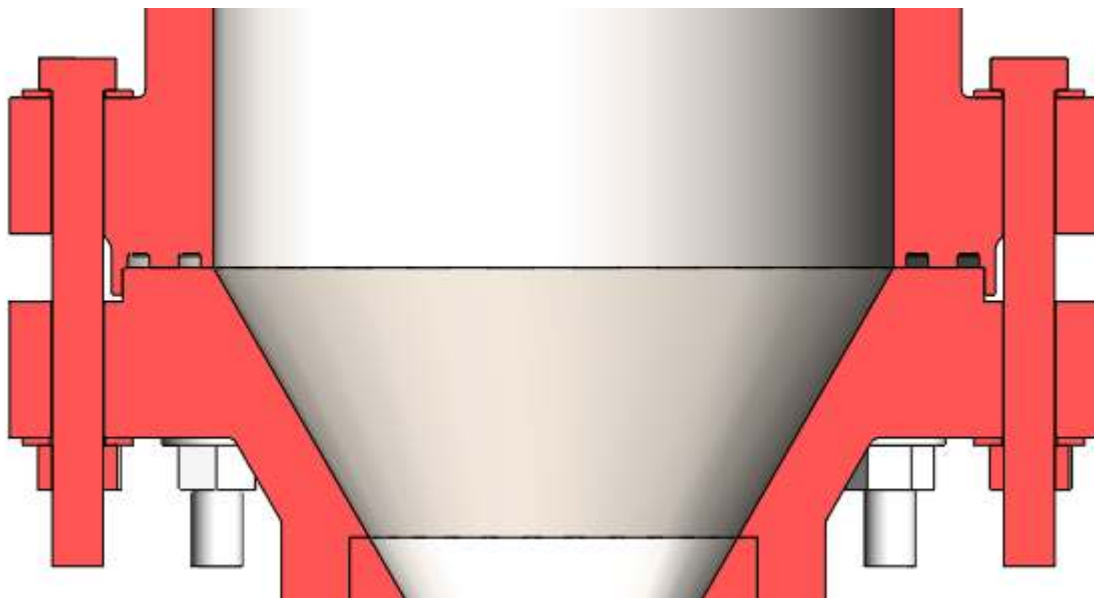


Figure 11: Bolted flange joint section

1" thick, integrally machined flanges are selected for their strength and stiffness. This prevents flexure and leakage during service. The robustness aids in surviving shock loads during hard starts. By machining the flanges into the parts, welding is not required and thus heat-affected zones and weld fatigue need not be considered. Table 5 is a summary of flange shear strength calculations. Detailed calculations are included in Appendix B.

Table 5: Bolted flange shear strength FoS

System Performance Case	Bolted flange shear strength FoS
Nominal 3 s burn	42.8
Hard start	11.0

Two Viton (KFM) O-rings are placed in series for redundancy in sealing. This arrangement was used by Youngblood [8] and by Austin and Heister [9] in their test engines successfully. O-ring sizes of #252 and #258 are selected as they comfortably fit within the limited sealing region. O-ring grooves are designed as per the Parker O-Ring Handbook [24] for face seal O-rings.

The O-ring grooves are placed on the combustion chamber for reusability. As the nozzle must be reusable after being ejected in a hard start, resurfacing a flat sealing face is significantly easier than repairing O-ring grooves.

Viton O-rings have a maximum continuous service temperature of 400 ° F [14] but this is alleviated by the short 3 s combustion and installation of new O-rings between burns. O-rings are considered consumable in this design as specialty high temperature O-rings are orders of magnitude more expensive [14]. PTFE O-rings have been shown to seal rocket engine burns [8] but their higher continuous service temperature is deemed unnecessary as they would still need to be replaced between burns. Additionally, PTFE O-rings are substantially more expensive than Viton O-rings and are significantly stiffer. Stiffer O-ring materials are less tolerant of surface imperfections on sealing faces.

For alignment during assembly, a circular lip is added to the combustion chamber sides of the joints. This feature aligns the axis of the combustion chamber and the axis of the nozzle. This prevents misalignment. Misalignment increases the risks of seal failure and part failure through hot spots created by sharp edges in the wetted surface.

The same type of bolt is universally applied across the engine for modularity and commonization of fasteners. 3/8"-16 grade 5 fully threaded 3.5" long bolts and standard height grade 5 nuts are the standard fastener in this design. Through hardened flat washers are used on both ends of the joint for load distribution and predictable torque characteristics. 3/8"-16 grade 5 bolts are selected such that the joint is practical for regular service while minimizing the diameter of the flange to reduce part weight

and material cost. Fully threaded 3.5" length fasteners are selected such that the same fasteners may be used in all the bolted flanges in this system.

Bolt patterns are selected in multiples of 4 for flange symmetry. The nozzle flange joint uses 8 bolts. The bolt preload is sized such that the joint will nominally fail at 1673 PSI. Table 6 is a summary of the bolt pattern calculations. Detailed calculations are included in Appendix B. As the injector flange joint is not designed to fail, a 16-bolt pattern with the same total joint preload is selected.

Table 6: Nozzle flange bolt calculation summary

Total Axial Bolt Preload	Preload % Yield Strength	2000 PSI Hard Start % UTS
41070 lbf	72.5 %	108.7 %

4.0 Conclusions

A robust and economical test platform for preliminary testing and characterization of a pintle injector is designed. The engine is designed to sustain 3 s burns and is sufficiently robust to be reusable after hard starts. Internal engine geometry is determined by RPA for a 5" inner diameter, 300 PSI NOS-ethanol engine. The engine consists of a combustion chamber, nozzle, and nozzle throat insert. The combustion chamber and nozzle are made of AISI 1020. The replaceable nozzle throat insert is made of AISI 304. Thermal analysis with Ansys FEA and RPA thermal parameters confirm the suitability of AISI 1020 for the combustion chamber and the nozzle. Bolted flange joints are used for assembly and Viton O-rings for sealing. The bolt pattern used in the nozzle flange joint is sized to nominally fail at 1673 PSI.

5.0 Next Steps

Next steps in this project include manufacturing, assembly, testing, and bolt torque preload determination. After design finalization, the design will be manufactured. The design will be assembled and hydrostatically tested to verify design calculations, particularly the actual failure pressure of the nozzle flange joint. Further research and testing are required to determine the correlation between bolt torque and axial preload for this bolted flange joint configuration; this work will increase consistency and accuracy in bolt preloading to achieve the desired failure pressure.

Once pintle injector design is finalized, an appropriate injector flange will be designed and manufactured to mount the pintle injector to the engine. Suitable fixtures must also be designed to attach the engine to the test stand.

To refine thermal models, a detailed temperature measurement plan must be specified for the static fire test campaign. Accurate thermal data and models will permit the design of light, optimized flight engines, and support later cooling system development as the engine design matures.

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Appendix A: RPA Results

Results are generated with Rocket Propulsion Analysis (RPA) software [25].

Thrust and mass flow rates

Chamber thrust (vac): 2.82366 kN

Specific impulse (vac): 239.94434 s

Chamber thrust (opt): 2.47929 kN

Specific impulse (opt): 210.68080 s

Total mass flow rate: 1.20000 kg/s

Oxidizer mass flow rate: 1.00000 kg/s

Fuel mass flow rate: 0.20000 kg/s

Geometry of thrust chamber with conical nozzle

$D_c = 125.06$ mm $b = 30.00$ deg

$R_2 = 0.34$ mm $R_1 = 0.02$ mm

$L^* = 1500.00$ mm

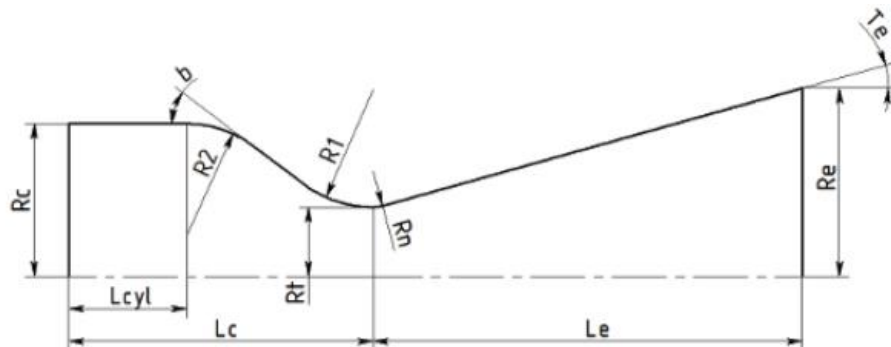
$L_c = 150.60$ mm $L_{cyl} = 71.07$ mm

$D_t = 33.34$ mm

$R_n = 0.02$ mm $T_e = 15.00$ deg

$L_e = 58.67$ mm

$D_e = 64.77$ mm



Results of thermal analysis

#	Location, mm	Radius, mm	Conv. heat coeff., k W/m^2-K	Conv. heat flux, kW/m^2	Rad. heat flux, kW/m^2	Total heat flux, kW/m^2	Twg, K	Twi, K	Twc, K	Tc, K	pC, MPa	wC, m/s	rho, kg/m^3	Comments
#	0.0000	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
4.4400	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
8.8800	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
13.3300	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
17.7700	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
22.2100	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
26.6500	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
31.0900	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
35.5400	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
39.9800	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
44.4200	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
48.8600	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
53.3000	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
57.7500	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
62.1900	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
66.6300	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
71.0700	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
75.5100	62.5300	62.5300	0.4325	560.1651	0.0000	560.1651	1876.0700	1876.0700	1874.5800	0.0000	0.0000	0.0000	0.0000	radiative cooling
80.0000	59.8000	59.8000	0.4729	598.6802	0.0000	598.6802	1907.9200	1907.9200	1906.0000	0.0000	0.0000	0.0000	0.0000	radiative cooling
84.4400	57.0300	57.0300	0.5222	641.9589	0.0000	641.9589	1942.0100	1942.0100	1939.5500	0.0000	0.0000	0.0000	0.0000	radiative cooling
88.8800	54.2500	54.2500	0.5779	689.7350	0.0000	689.7350	1977.8000	1977.8000	1974.5700	0.0000	0.0000	0.0000	0.0000	radiative cooling
93.3200	51.4700	51.4700	0.6427	742.8082	0.0000	742.8082	2015.5600	2015.5600	2011.6100	0.0000	0.0000	0.0000	0.0000	radiative cooling
97.7600	48.6900	48.6900	0.7185	801.7800	0.0000	801.7800	2055.3400	2055.3400	2050.4000	0.0000	0.0000	0.0000	0.0000	radiative cooling
102.2000	45.9100	45.9100	0.8106	869.4871	0.0000	869.4871	2098.5600	2098.5600	2092.3800	0.0000	0.0000	0.0000	0.0000	radiative cooling
106.6400	43.1300	43.1300	0.9214	945.9352	0.0000	945.9352	2144.6100	2144.6100	2136.9300	0.0000	0.0000	0.0000	0.0000	radiative cooling
111.0800	40.3500	40.3500	1.0568	1032.8378	0.0000	1032.8378	2193.9300	2193.9300	2184.4000	0.0000	0.0000	0.0000	0.0000	radiative cooling
115.5200	37.5700	37.5700	1.2241	1131.7589	0.0000	1131.7589	2246.7000	2246.7000	2234.9300	0.0000	0.0000	0.0000	0.0000	radiative cooling
119.9600	34.7900	34.7900	1.4345	1244.9740	0.0000	1244.9740	2303.3600	2303.3600	2288.8400	0.0000	0.0000	0.0000	0.0000	radiative cooling
124.4000	32.0100	32.0100	1.7050	1375.4913	0.0000	1375.4913	2364.5200	2364.5200	2346.6000	0.0000	0.0000	0.0000	0.0000	radiative cooling
128.8400	29.2400	29.2400	2.0552	1524.5330	0.0000	1524.5330	2429.8200	2429.8200	2407.7400	0.0000	0.0000	0.0000	0.0000	radiative cooling
133.2800	26.4600	26.4600	2.5215	1694.3586	0.0000	1694.3586	2499.2800	2499.2800	2472.1600	0.0000	0.0000	0.0000	0.0000	radiative cooling
137.7200	23.6800	23.6800	3.1417	1883.9660	0.0000	1883.9660	2571.5900	2571.5900	2538.5900	0.0000	0.0000	0.0000	0.0000	radiative cooling
142.1600	20.9000	20.9000	3.9655	2089.5602	0.0000	2089.5602	2644.4600	2644.4600	2605.1800	0.0000	0.0000	0.0000	0.0000	radiative cooling
146.6000	18.1200	18.1200	5.0012	2295.0390	0.0000	2295.0390	2712.1900	2712.1900	2667.2300	0.0000	0.0000	0.0000	0.0000	radiative cooling
151.0400	15.3400	15.3400	6.2974	2346.7542	0.0000	2346.7542	2728.2300	2728.2300	2681.0900	0.0000	0.0000	0.0000	0.0000	radiative cooling
155.4800	12.5600	12.5600	7.8822	2346.7542	0.0000	2346.7542	2596.1200	2596.1200	2589.1200	0.0000	0.0000	0.0000	0.0000	radiative cooling
159.9200	9.7800	9.7800	9.8800	1546.2523	0.0000	1546.2523	2385.1200	2385.1200	2385.1200	0.0000	0.0000	0.0000	0.0000	radiative cooling
164.3600	7.0000	7.0000	12.8800	1394.0307	0.0000	1394.0307	2375.3600	2375.3600	2385.1200	0.0000	0.0000	0.0000	0.0000	radiative cooling
168.8000	4.2200	4.2200	17.1515	1184.4281	0.0000	1184.4281	2274.7800	2274.7800	2259.5500	0.0000	0.0000	0.0000	0.0000	radiative cooling
173.2400	1.4400	1.4400	24.5600	1011.6171	0.0000	1011.6171	2184.1900	2184.1900	2173.6900	0.0000	0.0000	0.0000	0.0000	radiative cooling
177.6800	0.0000	0.0000	33.3900	860.3781	0.0000	860.3781	2094.4100	2094.4100	2086.8800	0.0000	0.0000	0.0000	0.0000	radiative cooling
#														

Appendix B: Calculations

Material properties are taken from Granta EduPack 2020 [26].

Bolt tensile yield strength, proof load, and ultimate tensile strength is taken from Portland Bolt [27].

Fastener tensile stress area is taken from Engineers Edge [28].

Combustion Chamber Calculations

Calculations are performed using thick wall pressure vessel formulas and the von-Mises yield stress criterion [29].

combustion chamber				
thick-wall pressure vessel and von-mises yield criterion				
condition	nominal 3 s burn		condition	hard start
failure mode	wall yield		failure mode	wall yield
internal radius (in)	2.5		internal radius (in)	2.5
external radius (in)	3		external radius (in)	3
internal pressure (psi)	315		internal pressure (psi)	2015
external pressure (psi)	15		external pressure (psi)	15
axial stress (psi)	667		axial stress (psi)	4530
internal hoop stress (psi)	1649		internal hoop stress (psi)	11076
internal radial stress (psi)	-315		internal radial stress (psi)	-2015
internal von-mises stress (psi)	1701		internal von-mises stress (psi)	11337
external hoop stress (psi)	1349		external hoop stress (psi)	9076
external radial stress (psi)	-15		external radial stress (psi)	-15
external von-mises stress (psi)	1181		external von-mises stress (psi)	7873
material	aisi 1020, annealed, 502 c		material	aisi 1020, annealed, 19.9 c
tensile yield strength (psi)	23206		tensile yield strength (psi)	39595
internal fos	13.6		internal fos	3.5
external fos	19.7		external fos	5.0

Insert Shear-Through of Retaining Flange Calculations

insert shear-through				
average shear stress				
condition	nominal 3 s burn		condition	hard start
failure mode	flange shear		failure mode	flange shear
insert diameter (in)	3		insert diameter (in)	3
flange thickness (in)	1		flange thickness (in)	1
pressure (psi)	300		pressure (psi)	2000
shear stress (psi)	225		shear stress (psi)	1500
material	aisi 1020, annealed, 502 c		material	aisi 1020, annealed, 19.9 c
tensile yield strength (psi)	23206		tensile yield strength (psi)	39595
shear yield strength (psi)	13390		shear yield strength (psi)	22846
fos	59.5		fos	15.2

Retaining Flange Bolt Calculations

retaining flange bolt	
tensile bolt failure	
condition	hard start
insert diameter (in)	3
pressure (psi)	2000
bolt (ct)	8
bolt diameter (in)	0.375
bolt tensile area (in^2)	0.077
bolt grade	5
ultimate tensile strength (psi)	120000
tensile yield strength (psi)	92000
total preload (lbf)	14137
preload per bolt (lbf)	1767
hard start load (lbf)	28274
preload % yield	24.9%
hard start % ultimate	38.2%

Bolted Flange Joint Shear Strength Calculations

bolted flange joint shear				
average shear stress				
condition	nominal 3 s burn		condition	hard start
failure mode	flange shear		failure mode	flange shear
inner diameter (in)	5			5
shear diameter (in)	6		shear diameter (in)	6
flange thickness (in)	1		flange thickness (in)	1
pressure (psi)	300		pressure (psi)	2000
shear stress (psi)	312.5		shear stress (psi)	2083
material	aisi 1020, annealed, 502 c		material	aisi 1020, annealed, 19.9 c
tensile yield strength (psi)	23206		tensile yield strength (psi)	39595
shear yield strength (psi)	13390		shear yield strength (psi)	22846
fos	42.8		fos	11.0

Bolted Flange Joint Bolt Calculations

bolted flange joint bolt	
tensile bolt failure	
condition	hard start
inner diameter (in)	5
pressure (psi)	2000
bolt (ct)	8
bolt diameter (in)	0.375
bolt tensile area (in^2)	0.077
bolt grade	5
ultimate tensile strength (psi)	120000
tensile yield strength (psi)	92000
seal preload (lbf)	1800
total preload (lbf)	41070
preload per bolt (lbf)	5134
hard start load (lbf)	80340
preload % yield	72.5%
hard start % ultimate	108.7%
failure pressure (psi)	1673